Experiment 25: Decision Tree for Iris Flower Classification

Aim: To classify iris flowers into three species using a Decision Tree classifier.

Algorithm:

1. Import the Iris dataset.
2. Split the dataset into features and labels.
3. Train the Decision Tree classifier.
4. Get flower measurements from the user.
5. Predict and display the species.

Code:

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier

iris = load_iris()

X = iris.data
y = iris.target

model = DecisionTreeClassifier()
model.fit(X, y)

sl = float(input("Sepal length: "))
sw = float(input("Sepal width: "))
pl = float(input("Petal length: "))
pw = float(input("Petal width: "))

result = model.predict([[sl, sw, pl, pw]])

print("Predicted Species:", iris.target_names[result[0]])
```

Input:

Sepal length: 5.1

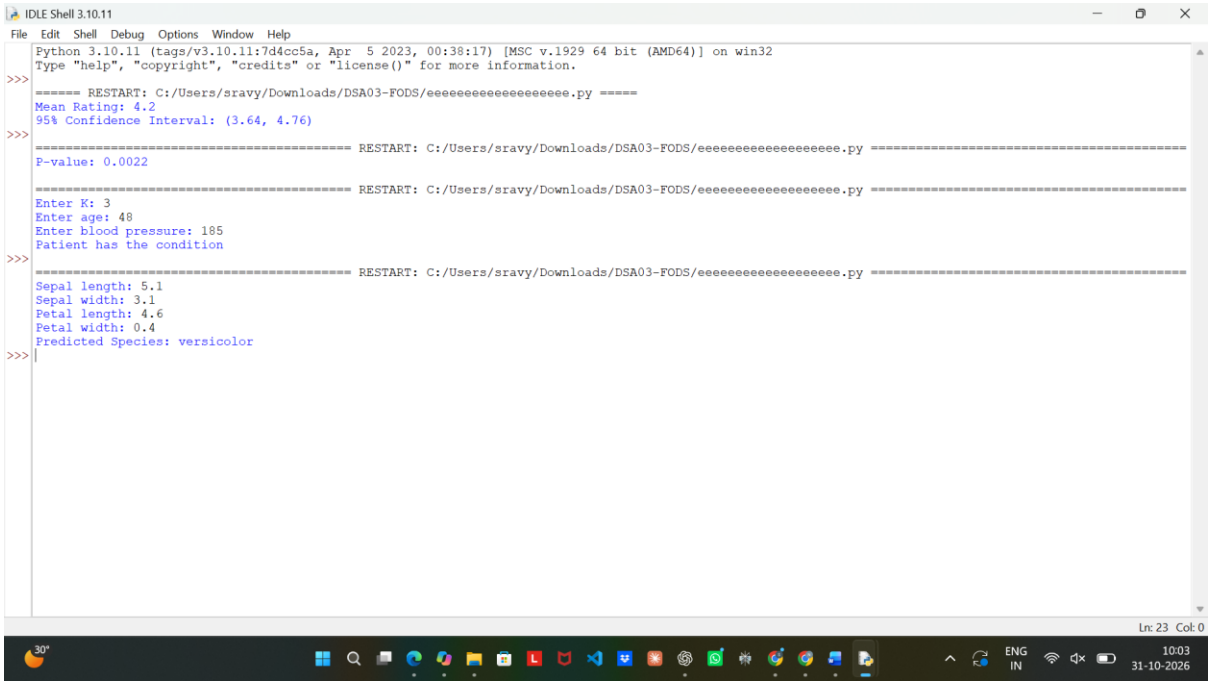
Sepal width: 3.5

Petal length: 1.4

Petal width: 0.2

Output:

Predicted Species: setosa



Result: The Decision Tree classifier successfully predicted the iris flower species.